The Marsh That Holds Its Breath

Student copy (project or hand out)

1 A wetland is an economy of patience. It runs on what it refuses to release. Drainage hurries; the intact marsh delays, and almost everything that makes a wetland matter follows from that single difference in pace. Water that would race downstream is held and let go slowly. Carbon that would burn off into the air is locked under standing water instead. The marsh is small on a map and enormous in what it keeps.

2 The keeping begins millimeters below the surface. Within a few millimeters of the waterline, heterotrophic bacteria have already used up the available oxygen, and the soil turns anaerobic, so decomposition nearly stops and dead plant matter accumulates instead of vanishing. That stalled rot is why wetlands, on only 5 to 8 percent of the planet's land, hold 20 to 30 percent of the carbon in the world's soils; peatlands alone, on 3 percent of ice-free land, store more than a quarter of it. Freshwater wetlands in the continental United States hold roughly 11.52 petagrams of carbon, most of it deeper than thirty centimeters, packed away below the reach of ordinary air.

3 On top of this slow chemistry sits a crowd. Around 40 percent of the world's plant and animal species depend on wetlands, including 30 percent of all known fish; over 100,000 freshwater species have been catalogued in them, and roughly 200 new ones are described each year. When floodplains flood on schedule, fish move in to spawn and the young feed in the nutrient-rich shallows before leaving for open water; the cast includes barramundi, tarpon, milkfish, and several kinds of bream. The plants that root here breathe through aerenchyma, porous channels that carry air down into airless mud, and stand on shallow roots, and seal their leaves with wax, and run hollow stems up to the light. In a good year that vegetation can lay down more than 6,000 grams of growth per square meter, far more than its animals can eat, so the surplus settles into the sediment as detritus and feeds the slow accounting below.

4 The marsh is a chemist that never finishes a reaction. Oxygen at the surface gives way to anaerobic conditions below, and that redox gradient decides, layer by layer, what form nitrogen, phosphorus, sulfur, and carbon will take and which ones the microbes can use. In the low-oxygen zone, denitrifying bacteria convert nitrate into nitrogen gas and vent it harmlessly to the sky; Europe's wetlands strip out roughly 1,092 kilotons of nitrogen this way each year. The same water it cleans, it also stores and rations: a wetland shaves peak river flow by an average of 13.8 percent for every kilometer of channel, with some reaches cutting more than 30 percent, while what soaks in recharges the aquifers and keeps streams running between rains.

5 Cut the standing water and the whole arrangement reverses. Air reaches the buried peat, the held breath lets go, and the bank that took centuries to fill begins to pay itself out. Exposed organic soil oxidizes and sinks, on the order of 4 centimeters a year at first and 2 a year after, until decades have eaten away

something like 121 centimeters of ground. The carbon that was a sink becomes a source. Nitrogen and phosphorus that the flooded soil had pinned in place wash downstream into algal blooms and fish kills, and a watershed that once shaved its floods can see peaks rise by as much as 80 percent. During Hurricane Sandy, the wetlands still standing on the East Coast absorbed an estimated 625 million dollars in flood damage that the houses behind them never had to.

6 What a drained marsh stops doing is harder to see than what it stops being. The water keeps arriving on its old schedule. The fish still come looking for the shallow water that timed their spawning to the flood, and the timing is gone.

AP-style multiple choice:

1. In the first paragraph, the writer opens by calling a wetland 'an economy of patience' and stating that 'drainage hurries; the intact marsh delays' primarily to

A) argue that human drainage of marshes should be slowed or halted to protect the climate
B) define the technical vocabulary of wetland science before the mechanisms are explained
C) establish the single contrast in pace that the rest of the opener will trace through carbon, water, and species
D) introduce the financial value of wetlands that the closing paragraphs will calculate in dollars

2. In the fifth paragraph, the description of the buried peat as a 'held breath' that 'lets go' once air reaches it most directly serves to

A) exaggerate the speed of soil collapse beyond what the cited subsidence rates support
B) compare the wetland to a human patient in order to argue for its protection
C) signal the writer's anger at the people responsible for draining the wetland
D) render an invisible chemical reversal as a physical, living event the reader can picture

Discussion questions:

1. The opener describes the same wetland twice: once as a system that holds water, carbon, nitrogen, and life in place, and once as a system unravelling after it is drained. How does the writer use one controlling contrast (delay against release) to connect mechanisms as different as bacterial respiration, fish spawning, and flood control?

2. This passage takes a clear position on what a wetland is worth without ever telling the reader to feel a certain way or do a certain thing. Where does the writing let facts carry the weight that an opinion piece would carry with adjectives or a call to action, and what does the writer give up by staying observational?

Writing prompt (Homework or extended in-class, Q2 rhetorical analysis):

The following passage describes how a wetland sustains itself and what happens when one is drained. Read the passage carefully. Then, in a well-written essay, analyze the rhetorical choices the writer makes to convey the marsh as a system whose worth lies in what it delays and holds rather than in what it does. In your response you should: respond to the prompt with a defensible thesis that analyzes the writer's rhetorical choices; select and use evidence from the passage to support your line of reasoning; explain how the evidence supports your reasoning; and demonstrate an understanding of the passage's rhetorical situation. Consider attending to the controlling contrast between delay and release, the use of accumulating technical detail and exact figures, and figurative touches such as the marsh as a 'chemist that never finishes a reaction' and the drained peat as a 'held breath' that lets go.